

TOPOLOGY OF ENTANGLEMENT

Persistent Homology as a Probe of the SYK / JT Gravity Correspondence

*A Complete Technical Record: First Principles, Construction, Results, and the Structure
Theorem*

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Repository: `github.com/ArjunParadkar/ER-EPR`

Record current through Rung 4 – commits `3ee3035` and `a2204e5`

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How to read this document

Parts II and III are self-contained derivations from base definitions; a reader who knows one field and not the other can skip the half they know. Part V is the experimental record and is the load-bearing section. Parts VIII and IX are the honesty apparatus — they exist because the central methodological finding of this project is that agreement between two calculations is evidence of nothing unless the calculations are independent. Every numerical claim in this document is tagged PROVEN, MEASURED, IMPORTED, or CONJECTURED.

Part o — Executive Summary and Status Board

The question was whether the persistent homology barcode of an entangled quantum state is holographically dual to the barcode of its gravitational geometry. The answer, at the scales we can reach, is a precise and falsifiable no — and the reason why turned out to be a theorem worth more than the yes would have been.

o.1 The one-paragraph version

The ER=EPR conjecture proposes that entanglement between two quantum systems is the same thing as a geometric wormhole connecting their duals. Existing tests of this idea compress entanglement into a single number — an entropy — and check that the number matches a single geometric quantity, an area. We asked a stronger question: does the entire multi-scale topological signature of the entangled state match the entire multi-scale topological signature of the geometry? Persistent homology gives a rigorous language for that question, because a barcode is a functor and two functors can be compared by a natural transformation. We built the machinery, constructed the transformation ϕ , and tested it on the cleanest available holographic pair: the Sachdev–Ye–Kitaev model and Jackiw–Teitelboim gravity. The construction fails to be a natural isomorphism at every system size we can compute. But in proving why it fails, we obtained a structure theorem: for any monotone circulant filtration built on a discretized thermal circle, the entire persistence module collapses onto two scalar functionals of the correlator. That collapse is the obstruction. It is exact, it is provable, and it tells you precisely what a construction would have to do to succeed.

o.2 Status board

Rung	Content	Status	Verdict
1	Two-qubit Bell state proof of concept	Complete	Pipeline validated. NOT a physics result — agreement was construction-guaranteed.
2	Thermofield double, temperature sweep	Complete	Same tautology as Rung 1. RT relation was an input, not a discovery.
3	Full SYK vs JT barcode comparison	Complete	First genuine topological content. Exposed the index mismatch; forced τ -indexing on both sides.
4	Finite-N corrections, systematic deviations	Committed 3ee3035, a2204e5	HONEST NEGATIVE. Shape failure of the Schwarzian profile at accessible N, established independent of calibration.
5	Natural isomorphism theorem, categorical	In progress — Naman	Structure theorem and corollary drafted. Functoriality proof outstanding.

o.3 The four headline results

Result 1 — The tautology diagnosis (methodological)

Rungs 1 and 2 produced a Wasserstein distance of exactly zero between the boundary barcode and the bulk barcode. This looked like a spectacular confirmation. It was not a result at all. ϕ had been defined as the ratio of two quantities that were themselves defined to be equal, so the vanishing distance was algebraically guaranteed before any physics entered. The Ryu–Takayanagi relation, which we believed we were testing, was in fact an input to the construction. Identifying this — and quarantining it before it was reported — is the single most important methodological event in the project.

Result 2 — The index mismatch (structural)

The SYK model is naturally indexed by fermion flavor $i \in \{1, \dots, N\}$. JT gravity is naturally indexed by radial position in the bulk. These two index sets are not related to each other by the holographic dictionary, and any comparison that silently identifies them is comparing incommensurable objects. The resolution is that both sides must be indexed by Euclidean time τ on the thermal circle. This is not a technicality; it changes what object is being built on each side.

Result 3 — The structure theorem (mathematical, PROVEN)

On a discretized thermal circle with $N_\tau = 16$ points, the correlator-induced distance depends only on the cyclic separation $|i - j| \bmod N_\tau$. The Vietoris–Rips filtration is therefore a filtration of circulant graphs, and by the Adamaszek–Adams classification the persistent homology is a function of the ordered separation distances alone. The entire barcode — every bar, every birth, every death — reduces to two scalar functionals of the correlator, $\hat{G}(1)$ and $\hat{G}(8)$, weighted approximately 23:1. There is no additional topological information in the boundary data. This is a hard structural ceiling, not an artifact of resolution.

Result 4 — The finite-N negative (physical, MEASURED)

At every system size accessible to exact diagonalization ($N \leq 18$ Majoranas, Hilbert space dimension $2^9 = 512$), the calibration constant C_{fit} relating boundary and bulk barcodes is not a constant. It depends on β with spreads of 29–158 % across temperature at fixed N , and the ratio $C_{\text{fit}}(\hat{G}(1))/C_{\text{fit}}(\hat{G}(8))$ ranges from 1.47 to 7.81. A single constant cannot simultaneously satisfy both channels. Separately and more strongly: the shape of the Schwarzian profile fails to match at accessible N , and this shape failure is established independent of any calibration choice — no rescaling can repair it.

What this means, stated without spin

ϕ is not a natural isomorphism at $N \leq 18$. We have not shown it fails in the large- N limit, and we do not claim to have. What we have is an exact criterion — stated in Part VII — for when a construction of this type can be a natural isomorphism, together with a measurement showing the criterion is violated in the regime we can access. That is a falsifiable, checkable, extendable result. A tautological yes would have been worth nothing.

0.4 What is imported rather than measured

The large- N Maldacena–Stanford dictionary — the map relating SYK correlator parameters to JT gravity geometric quantities — is used throughout as an IMPORTED CALIBRATION. We did not derive it and we did not measure it. It is taken from the literature and applied. This distinction is maintained explicitly in the ledger in

Part IX, because conflating an imported relation with a measured one is exactly how the Rung 1–2 tautology arose in the first place.

0.5 Outstanding items

- Push access to github.com/ArjunParadkar/ER-EPR is blocked by a 403 — Naman is not listed as a collaborator, and all Rung 4 commits currently exist only on a local machine. This is the highest-priority loose end in the project and is a single-point-of-failure risk.
- Collaborator handoff report for Arjun, covering the Rung 4 closure and the structure-theorem reframe.
- Rung 5 functoriality proof — the naturality square must be shown to commute for all morphisms in the filtration poset, not merely checked pointwise.
- Paper integration: Rungs 1–4 drafted with embedded figures; Part II skeleton in place.

Part I — Motivation: Why This Question Exists

Everything in this project descends from a single suspicion: that the standard tests of ER=EPR throw away almost all of the information in the state before comparing anything.

1.1 The ER=EPR conjecture

In 1935 Einstein and Rosen described a solution to the field equations in which two distant regions of spacetime are joined by a bridge. In the same year Einstein, Podolsky and Rosen described a pair of particles whose measurement outcomes are correlated in a way no local description can reproduce. For nearly eighty years these were regarded as unrelated papers that happened to share an author.

Maldacena and Susskind proposed in 2013 that they are the same paper. The claim, written compactly, is:

$$ER = EPR \quad (\text{Maldacena–Susskind, 2013})$$

Two systems that are entangled are connected by a geometric bridge in the dual gravitational description, and the amount and structure of the entanglement determines the geometry of the bridge. In the sharpest version of the statement, an eternal black hole in anti-de Sitter space with two asymptotic boundaries is dual to a specific entangled state — the thermofield double — living on two copies of a boundary quantum theory. The wormhole is not a metaphor for the entanglement. It is the entanglement, seen in the other description.

1.2 The entropy bottleneck

The standard quantitative test of this correspondence is the Ryu–Takayanagi relation and its covariant generalizations. It says that the entanglement entropy of a boundary region A is given by the area of a minimal surface γ_A in the bulk that is anchored to the boundary of A:

$$S_A = \text{Area}(\gamma_A) / 4G_N \quad (\text{Ryu–Takayanagi})$$

This is a beautiful and heavily verified relation. It is also, viewed as a test, extremely lossy. The left-hand side is one real number. The right-hand side is one real number. An entangled quantum state of N fermions is an object with exponentially many parameters; its reduced density matrix has an entire spectrum. Compressing all of that into a single scalar and matching it against a single geometric scalar is a weak constraint. Many inequivalent states share an entropy.

The motivating gap

If ER=EPR is true in the strong sense — if entanglement structure IS geometric structure — then the correspondence should survive being tested with a richer invariant than a single number. Nobody had asked whether the multi-scale topological signature of the entangled state matches the multi-scale topological signature of the geometry, because until recently there was no clean way to ask.

1.3 Why persistent homology is the right tool

Persistent homology takes a dataset equipped with a notion of distance and returns a barcode: a multiset of intervals recording, for every homological degree, the range of scales over which each topological feature exists. It has three properties that make it the correct instrument here.

It is multi-scale by construction

A barcode does not commit to a single resolution. It records what the data looks like at every resolution simultaneously, and which features survive across resolutions. Entanglement in a many-body system is itself organized by scale — short-range correlations, long-range correlations, and the crossover between them. An invariant that respects scale is structurally matched to the physics.

It is stable

The stability theorem guarantees that a small perturbation of the input distances produces a small perturbation of the barcode, measured in bottleneck distance. This is essential: numerical correlators carry sampling noise and finite-size error, and an invariant that amplified those errors would be useless. Persistent homology is provably Lipschitz in the input.

It is a functor, and functors can be compared properly

This is the decisive point and the reason the project is stated categorically. A persistence module is not merely a list of intervals; it is a functor from the poset of scales into vector spaces. When we have a functor on the boundary side and a functor on the bulk side, the mathematically correct question is not 'are these barcodes close?' but 'is there a natural transformation between these functors, and is it invertible?' Naturality is strictly stronger than pointwise agreement: it demands that the comparison commute with every scale-change morphism, not just that it hold at each scale separately. A construction can agree at every scale and still fail to be natural.

Why naturality is the real content

Two barcodes being numerically equal is a statement about outputs. A natural isomorphism is a statement about the entire structure that produced them, uniformly across all scales. It is the difference between two functions having the same values on a sample and two functions being the same function. This distinction is what elevates the project from a numerical comparison to a theorem.

1.4 Why SYK and JT gravity

The correspondence needs a testbed where both sides are actually computable. Most holographic pairs fail this test: the boundary theory is strongly coupled and intractable, or the bulk is higher-dimensional and its geometry is not solvable in closed form. The SYK/JT pair is the rare case where both sides are under control.

- The SYK model is a quantum-mechanical model of N Majorana fermions with random all-to-all four-fermion couplings. It is exactly solvable at large N because the disorder average is dominated by a single class of diagrams, and it is exactly diagonalizable at small N because the Hilbert space is only $2^{N/2}$ -dimensional.

- JT gravity is two-dimensional dilaton gravity. In two dimensions the Einstein tensor vanishes identically, so all of the dynamics lives on the boundary, and that boundary dynamics is a single well-understood action — the Schwarzian.
- Both sides are governed by the same Schwarzian theory in the appropriate limit. This is the content of the SYK/JT correspondence and it is why the pair is the standard laboratory for near-extremal holography.
- The eternal black hole in JT and the thermofield double state in SYK are an explicit, concrete instance of the ER=EPR pairing. This is exactly the object the conjecture is about.

1.5 The precise question this project asks

Assembling the above, the question is:

The central question
Let F_{bdy} be the persistence module obtained from the SYK thermofield double state at inverse temperature β , and let F_{bulk} be the persistence module obtained from the corresponding JT gravity eternal black hole geometry. Is there a natural transformation $\phi : F_{\text{bdy}} \Rightarrow F_{\text{bulk}}$, and is ϕ a natural isomorphism?

A yes would be a substantially stronger statement of ER=EPR than any entropy matching, because it would say the entire scale-resolved topological structure — not one number — is shared. A no, if accompanied by a proof of exactly what obstructs it, is a structural theorem about what holographic duality does and does not preserve. Both outcomes are publishable. Only one of them was available, and it was the second.

1.6 The five-rung strategy

The project was structured as a ladder, each rung a strictly harder test than the last, with the explicit rule that no rung is climbed until the one below it has been audited for construction-guaranteed agreement.

Rung	Object	What it tests	Why it must come first
1	Two entangled qubits, Bell state	Does the pipeline run end to end and produce sane barcodes?	The smallest system where entanglement exists at all. If it fails here, nothing above matters.
2	Thermofield double, varying β	Does temperature dependence propagate correctly through the pipeline?	Introduces the thermal circle, which is the geometric arena for everything above.
3	Full SYK vs JT barcodes	Is there genuine, non-trivial topological content to compare?	First rung where the answer is not fixed by construction.
4	Finite-N corrections	Do the deviations behave like quantum corrections should?	Distinguishes a real correspondence from an accidental one.
5	Natural isomorphism theorem	Is ϕ natural and invertible, provably?	The actual claim. Everything below is instrumentation.

The rule that made the ladder work was not the ordering. It was the audit condition attached to each rung: before a rung counts as climbed, we ask whether its agreement could have been produced by the construction alone. Rungs 1 and 2 failed that audit. Catching that is what made Rungs 3 and 4 worth running.

Part II — Mathematical First Principles

This part builds the entire mathematical apparatus from base definitions. Nothing is assumed beyond linear algebra and the notion of a topological space. A physicist who has never seen a chain complex should be able to read this straight through.

2.1 Simplicial complexes

The first task is to turn a set of points into something that has topology. A point cloud, as a set, has no interesting topology — it is a finite discrete space, and every finite discrete space looks the same. Topology has to be manufactured by deciding which points are close enough to be joined.

An abstract simplicial complex K on a finite vertex set V is a collection of subsets of V , called simplices, closed under taking subsets: if $\sigma \in K$ and $\tau \subseteq \sigma$, then $\tau \in K$. A simplex with $k+1$ vertices is a k -simplex. A 0-simplex is a vertex, a 1-simplex is an edge, a 2-simplex is a filled triangle, a 3-simplex is a filled tetrahedron.

The closure condition is what makes the object geometric: you cannot have a filled triangle without its three edges. Geometrically, K is a recipe for gluing together solid triangles, tetrahedra and their higher analogues along shared faces, and the resulting space is what we compute the topology of.

The Vietoris–Rips complex

Given a finite metric space (X, d) and a scale parameter $r \geq 0$, the Vietoris–Rips complex $VR(X, r)$ is defined by:

$$\sigma \subseteq X \text{ is a simplex of } VR(X, r) \iff d(x, y) \leq r \text{ for all } x, y \in \sigma$$

In words: a set of points spans a simplex exactly when all of them are pairwise within r . This is the clique complex of the graph whose edges are the pairs at distance at most r . It is the cheapest possible construction — it is determined entirely by the pairwise distance matrix, requires no ambient embedding, and needs no coordinates.

That last property is what makes it usable here. Our data is a correlation matrix. There is no ambient space in which SYK fermions live. Vietoris–Rips takes a matrix of pairwise numbers and returns a topological space, which is precisely the operation the project needs.

Monotonicity — the property everything depends on

If $r \leq r'$ then $VR(X, r) \subseteq VR(X, r')$. Increasing the scale can only add simplices, never remove them. This nesting is what turns a family of complexes into a filtration, and a filtration is what persistence is computed from. Every construction in this project must preserve monotonicity or the entire apparatus collapses.

2.2 Simplicial homology

Homology assigns to a space a sequence of vector spaces whose dimensions count holes of each dimension: H_0 counts connected components, H_1 counts loops that do not bound a disk, H_2 counts enclosed voids, and so on.

Chain groups

Fix a field k (we use $k = \mathbb{Z}/2$ throughout; over a field, homology is a vector space and the classification theorem below applies cleanly). The k -th chain group $C_k(K)$ is the k -vector space with basis the k -simplices of K . An element is a formal linear combination of k -simplices, called a k -chain.

The boundary map

The boundary operator $\partial_k : C_k \rightarrow C_{k-1}$ sends a simplex to the alternating sum of its codimension-one faces:

$$\partial_k [v_0, v_1, \dots, v_k] = \sum_{i=0}^k (-1)^i [v_0, \dots, \hat{v}_i, \dots, v_k]$$

where the hat denotes omission. The boundary of a filled triangle is its three edges; the boundary of an edge is its two endpoints.

The fundamental lemma

The single algebraic fact on which all of homology rests is:

$$\partial_{k-1} \circ \partial_k = 0 \quad (\text{the boundary of a boundary is zero})$$

Every face of a simplex is shared by exactly two codimension-one faces, and the alternating signs make those two contributions cancel. Consequently $\text{im } \partial_{k+1} \subseteq \ker \partial_k$, and the quotient is well defined.

Homology groups

$$H_k(K) = \ker \partial_k / \text{im } \partial_{k+1}$$

Elements of $\ker \partial_k$ are cycles — chains with no boundary, i.e. closed loops in degree 1. Elements of $\text{im } \partial_{k+1}$ are boundaries — cycles that already bound something filled in. The quotient counts cycles that are not boundaries: genuine holes. The dimension $\beta_k = \dim H_k$ is the k -th Betti number.

What we actually need

For this project only H_0 and H_1 carry information. H_0 measures how the correlator organizes the thermal circle into clusters as the scale grows. H_1 measures whether the thermal circle survives as a topological loop and at what range of scales — this is the wormhole-relevant feature, because the loop is the discretized Euclidean time circle whose behaviour encodes the thermal geometry. Higher degrees are computed but, as Part VI shows, are structurally determined by the same two scalars.

2.3 Filtrations and persistence modules

A single complex at a single scale r is not enough. Choosing r is choosing an answer: too small and every point is isolated, too large and everything is one contractible blob. The whole point of persistence is to refuse to choose.

A filtration is the nested family $\{VR(X, r)\}_{r \geq 0}$. Applying H_k to it, and using the fact that an inclusion of complexes induces a linear map on homology, produces for every $r \leq r'$ a map:

$$H_k(VR(X, r)) \rightarrow H_k(VR(X, r'))$$

These maps compose correctly and the identity goes to the identity. That is precisely the definition of a functor.

Persistence modules as functors

Regard $(\mathbb{R}, \leq)$ as a category: objects are real numbers, and there is exactly one morphism $r \rightarrow r'$ whenever $r \leq r'$. A persistence module is a functor

$$M : (\mathbb{R}, \leq) \rightarrow \text{Vect}_k$$

assigning a vector space $M(r)$ to each scale and a linear map $M(r \leq r') : M(r) \rightarrow M(r')$ to each morphism, functorially. The persistent homology of a filtration is exactly such a functor. This is not a reformulation for elegance — it is the reason the project can be stated as a theorem rather than a numerical comparison, because functors admit natural transformations and mere lists of numbers do not.

2.4 The structure theorem and barcodes

An interval module $I[b, d]$ is the persistence module with a one-dimensional vector space at every scale in $[b, d]$, zero elsewhere, and identity maps within the interval. It is the algebraic encoding of a feature that is born at b and dies at d .

Structure theorem (Crawley-Boevey; Gabriel for the finite case) — PROVEN

Every pointwise finite-dimensional persistence module over $\mathbb{R}$ decomposes as a direct sum of interval modules, and the decomposition is unique up to reordering. That is, $M \cong \bigoplus_{i \in I} I[b_i, d_i]$.

The multiset of intervals $\{[b_i, d_i]\}$ is the barcode. This theorem is what makes persistent homology a practical tool: it says that an infinite-dimensional-looking functor is completely described by a finite list of intervals, and that this list is a complete invariant. Two persistence modules are isomorphic if and only if their barcodes agree.

The equivalent picture is the persistence diagram: plot each bar as a point (b_i, d_i) in the plane above the diagonal. Long bars sit far from the diagonal and represent robust features; short bars hug the diagonal and are typically noise.

2.5 Distances between barcodes

To compare two barcodes we need a metric.

Bottleneck distance

Given diagrams D_1 and D_2 , augmented with the diagonal (so that points may be matched to 'nothing'), the bottleneck distance is:

$$d_B(D_1, D_2) = \inf_{\eta} \sup_{x \in D_1} \|x - \eta(x)\|_{\infty}$$

over all bijections η between the augmented diagrams. It is the worst single mismatch under the best possible matching.

Wasserstein distance

$$W_p(D_1, D_2) = (\inf_{\eta} \sum_{x \in D_1} \|x - \eta(x)\|_{\infty}^p)^{1/p}$$

This is the corresponding p -norm aggregate rather than the supremum. It is more sensitive to distributed disagreement than the bottleneck distance and is the primary comparison metric used in Rungs 1 through 4.

Interleaving distance and the isometry theorem

The intrinsically categorical distance is the interleaving distance. Modules M and N are ϵ -interleaved if there are natural transformations $f : M \Rightarrow N(\cdot + \epsilon)$ and $g : N \Rightarrow M(\cdot + \epsilon)$ whose composites are the internal shift maps by 2ϵ . The interleaving distance d_I is the infimum of such ϵ .

Isometry theorem — PROVEN

$d_I(M, N) = d_B(\text{barcode } M, \text{barcode } N)$. The abstract categorical distance and the concrete combinatorial matching distance are the same distance. This is what licenses reading a numerical bottleneck computation as a statement about functors — and it is why Wasserstein numbers in Rungs 1–4 have categorical meaning at all.

The stability theorem

$$d_B(Dgm(f), Dgm(g)) \leq \|f - g\|_\infty$$

Perturbing the filtration function by at most ϵ moves the diagram by at most ϵ in bottleneck distance. Persistent homology is 1-Lipschitz. Numerical noise in a correlator therefore cannot produce large spurious barcode differences — which means that when we do see a large difference, as in Rung 4, it is signal.

2.6 Category theory: what the theorem actually claims

This section defines the objects in the target statement of Rung 5.

Categories and functors

A category C consists of objects, morphisms between them, associative composition, and identity morphisms. A functor $F : C \rightarrow D$ assigns objects to objects and morphisms to morphisms, preserving composition and identities: $F(g \circ f) = F(g) \circ F(f)$ and $F(\text{id}) = \text{id}$.

Our two functors are: F_{bdy} , the persistence module of the SYK boundary data, and F_{bulk} , the persistence module of the JT gravity geometry. Both are functors $(\mathbb{R}, \leq) \rightarrow \text{Vect}_k$.

Natural transformations

Given functors $F, G : C \rightarrow D$, a natural transformation $\phi : F \Rightarrow G$ is a family of morphisms $\phi_X : F(X) \rightarrow G(X)$, one for each object X , such that for every morphism $f : X \rightarrow Y$ in C the following square commutes:

$$G(f) \circ \phi_X = \phi_Y \circ F(f) \quad (\text{the naturality square})$$

Naturality is the crucial and easily-underestimated condition. It says that the comparison is not made independently at each scale and then hoped to line up; it says the comparison is compatible with every scale-change map in the filtration. A family of isomorphisms that fails this condition is not a natural transformation at all, no matter how well it performs pointwise.

Natural isomorphism

ϕ is a natural isomorphism if every component ϕ_X is an isomorphism in D . Equivalently there is a natural transformation $\psi : G \Rightarrow F$ with $\psi \circ \phi = \text{id}_F$ and $\phi \circ \psi = \text{id}_G$. When such a ϕ exists, F and G are the same functor up to canonical identification.

The Rung 5 target, stated precisely

Construct $\phi : F_{\text{bdy}} \Rightarrow F_{\text{bulk}}$ explicitly; prove that every naturality square commutes (functoriality); prove that each component is invertible (invertibility, via the structure theorem). Then and only then is the sentence 'the entanglement barcode and the geometry barcode are the same object' a theorem rather than an observation.

2.7 Vietoris–Rips complexes of the circle: Adamaszek–Adams

This theorem is the mathematical engine of the entire project, and it was not anticipated at the outset. It is what converts the negative result into a structure theorem.

The thermal circle is a circle. Our data lives on a discretization of it. So the relevant question is: what does the Vietoris–Rips complex of a circle look like as the scale grows? Naively one expects it to look like a circle until the scale is large enough to fill it in, then become contractible. That is false, and the truth is much more rigid.

Adamaszek–Adams theorem — PROVEN

For the circle S^1 with geodesic metric normalized to circumference 1, the Vietoris–Rips complex passes through a strictly increasing sequence of odd-dimensional spheres as the scale grows: $VR(S^1; r) \simeq S^{\ell(2\ell+1)}$ for $\ell/(2\ell+1) < r < (\ell+1)/(2\ell+3)$, and is contractible for $r \geq 1/2$. It is never anything else.

Concretely: for $r < 1/3$ the complex is homotopy equivalent to S^1 . At $r = 1/3$ it jumps to S^3 . At $r = 2/5$ it jumps to S^5 . Then S^7 , and so on, before finally collapsing to a point. The homotopy type is a step function of the scale with completely determined jump locations.

Why this matters here

Three consequences, each load-bearing:

1. The H_1 bar of a circle-like point cloud has a birth determined by when the cycle first closes and a death determined by exactly when the complex jumps from S^1 to S^3 . Both endpoints are pinned by the theorem — they are not free parameters that the physics can adjust.
2. Because the jump thresholds are fixed rational numbers, the barcode of a circulant filtration is a function of the ordered distance data alone, with no additional freedom. This is the seed of the structure theorem in Part VI.
3. It explains, without any appeal to physics, why the barcodes we compute are so information-poor. It is not that our resolution is bad. It is that the object genuinely does not carry more information than this.

2.8 A necessary lemma: one-dimensional point clouds have trivial H_1

We need this to rule out a class of degenerate constructions, and the first two arguments we produced for it were wrong. The correct proof is recorded here; the incorrect ones are recorded in Part VIII.

Lemma — PROVEN

Let X be a finite set of points in $\mathbb{R}$ with the Euclidean metric. Then $H_1(VR(X, r)) = 0$ for every $r \geq 0$.

Proof. Fix r . The graph G_r with vertex set X and an edge between x and y whenever $|x - y| \leq r$ is an indifference graph (equivalently, a unit interval graph): its vertices are points on a line and adjacency is proximity within a fixed threshold. Indifference graphs are chordal — every cycle of length four or more has a chord. Chordal graphs are dismantlable, and the clique complex of a chordal graph is collapsible, hence contractible. Since $VR(X, r)$ is by definition the clique complex of G_r , it is contractible, so all reduced homology vanishes, in particular $H_1 = 0$. As r was arbitrary, the H_1 persistence module is identically zero. ■

Two arguments that do NOT prove this

We initially argued from curvature (that a one-dimensional cloud has no curvature to sustain a loop) and separately from triangle-filling (that triangles always fill in before a loop can close). Both are wrong. Curvature is not a simplicial notion and does not enter the definition of VR at all; the triangle-filling claim is not generally true and does not by itself imply vanishing H_1 . Only the chordality argument is a proof. This is recorded because the wrong arguments felt persuasive, which is exactly what makes them dangerous.

Consequence for the project

Any construction that produces essentially one-dimensional data has no H_1 content whatsoever, and therefore cannot be probing wormhole topology no matter what the numbers say. This lemma is the reason the τ -indexed circular construction is not optional — it is the minimal construction with any hope of non-trivial H_1 .

Part III — Physical First Principles

This part builds the physics from base definitions: what a Majorana fermion is, what the SYK model is, why it is solvable, what the Schwarzian is, what JT gravity is, and what the thermofield double state is. A mathematician who has never seen a path integral should be able to read this straight through.

3.1 Majorana fermions

A Majorana fermion is a fermionic degree of freedom that is its own antiparticle. Where a complex fermion has independent creation and annihilation operators, a Majorana operator satisfies $\chi^\dagger = \chi$. The algebra is the Clifford algebra:

$$\{\chi_i, \chi_j\} = \chi_i \chi_j + \chi_j \chi_i = \delta_{ij}$$

with the normalization convention $\chi_i^2 = 1/2$. For N Majorana modes the operators generate a Clifford algebra whose irreducible representation has dimension $2^{N/2}$. This is the entire Hilbert space, and it is the reason exact diagonalization is feasible at modest N : for $N = 18$ the Hilbert space has dimension $2^9 = 512$, which is a trivially small matrix to diagonalize completely.

Concretely, Majorana operators can be built from ordinary spin operators by a Jordan–Wigner transformation, which is how they are represented in the numerical implementation. Two Majoranas combine into one complex fermion; the factor of $2^{N/2}$ rather than 2^N is exactly this pairing.

3.2 The SYK model

The Sachdev–Ye–Kitaev model is N Majorana fermions with random, all-to-all, four-body interactions:

$$H = (1/4!) \sum_{i,j,k,l=1}^N J_{ijkl} \chi_i \chi_j \chi_k \chi_l$$

The couplings J_{ijkl} are totally antisymmetric in their indices and drawn independently from a Gaussian distribution with zero mean and variance:

$$\langle J_{ijkl}^2 \rangle = 3! J^2 / N^3$$

The N -scaling of the variance is not cosmetic — it is chosen precisely so that the model has a sensible large- N limit in which the energy is extensive. Change that scaling and the model either becomes free or becomes divergent.

Three properties that make SYK special

- No quasiparticles. The model is maximally chaotic: its out-of-time-order correlator grows at the Maldacena–Shenker–Stanford bound $\lambda_L = 2\pi/\beta$. Black holes saturate the same bound. This is the first concrete hint that SYK is describing a black hole.
- Solvable at large N . The disorder average is dominated by a single family of diagrams — the melons — and all other diagram topologies are suppressed by powers of $1/N$. This gives closed Schwinger–Dyson equations.

- An emergent reparametrization symmetry. At low energies the model becomes invariant under time reparametrizations, and the spontaneous and explicit breaking of that symmetry produces the Schwarzian action, which is the same action that governs JT gravity. This is the correspondence.

3.3 The large-N solution

Define the Euclidean two-point function, averaged over disorder and over flavor:

$$G(\tau_1, \tau_2) = (1/N) \sum_i \langle T \chi_i(\tau_1) \chi_i(\tau_2) \rangle$$

At large N the self-energy is given by the melon sum, which is a single term:

$$\Sigma(\tau) = J^2 G(\tau)^3$$

and the Dyson equation in frequency space closes the system:

$$G(i\omega)^{-1} = -i\omega - \Sigma(i\omega)$$

These two equations together are the Schwinger–Dyson equations of the model. They are solved numerically by iteration: guess G, compute Σ in time domain, transform, apply the Dyson equation, transform back, repeat to convergence. This is the core of Arjun's numerical implementation.

The conformal limit

At strong coupling, $\beta J \gg 1$, the $-i\omega$ term is negligible compared to the self-energy and can be dropped. The resulting equations have an exact scaling solution:

$$G_c(\tau) = b \cdot \text{sgn}(\tau) / |\tau|^{2\Delta}, \quad \Delta = 1/4, \quad \pi J^2 b^4 = 1/4$$

The dimension $\Delta = 1/4$ follows directly from the quartic interaction: the Dyson equation without the $-i\omega$ term forces $G * \Sigma = -1$, and with $\Sigma \propto G^3$ this fixes $4\Delta = 1$.

At finite temperature the conformal solution is obtained by mapping the line to the thermal circle of circumference β , which for a dimension- Δ operator gives:

$$G_\beta(\tau) = b \cdot \text{sgn}(\tau) \cdot [\pi / (\beta J \sin(\pi \tau / \beta))]^{2\Delta}$$

This is the central object of the whole project. It is a function on the Euclidean time circle, it depends on τ only through the separation, and it is symmetric under $\tau \rightarrow \beta - \tau$. Those three facts are precisely the hypotheses of the structure theorem in Part VI.

3.4 Reparametrization symmetry and the Schwarzian

The conformal equations, having dropped the $-i\omega$ term, are invariant under any smooth reparametrization $\tau \rightarrow f(\tau)$, acting as:

$$G(\tau_1, \tau_2) \rightarrow [f'(\tau_1) f'(\tau_2)]^{-\Delta} G(f(\tau_1), f(\tau_2))$$

This is an enormous symmetry — the full diffeomorphism group of the circle. The conformal solution is not invariant under all of it; it is invariant only under the $SL(2, \mathbb{R})$ subgroup. So the symmetry is spontaneously broken from $\text{Diff}(S^1)$ down to $SL(2, \mathbb{R})$, and the resulting Goldstone modes are the reparametrization modes f .

Crucially, the symmetry is also explicitly broken, because the $-i\omega$ term we dropped is not actually zero — it is merely small. That explicit breaking gives the Goldstone modes an action, and to leading order in the derivative expansion that action is uniquely fixed by the residual $SL(2, \mathbb{R})$ invariance to be the Schwarzian:

$$S[f] = -N \alpha_S / J \int d\tau \text{Sch}(f, \tau)$$

$$\text{Sch}(f, \tau) = f'''/f' - (3/2) (f''/f')^2$$

The Schwarzian derivative is the unique — up to normalization — differential expression annihilated exactly by Möbius transformations, which is why it and nothing else can appear here. The coefficient α_S is a pure number determined by the microscopic model.

Why the Schwarzian is the whole story

Every low-energy observable in SYK — the specific heat, the thermodynamics, the out-of-time-order correlator, and the correlation functions we feed into the persistence pipeline — is computed from this one action. The same action arises independently on the gravity side. The correspondence is the statement that these are the same theory.

3.5 JT gravity

General relativity in two dimensions is trivial: the Einstein tensor vanishes identically, so the metric has no local dynamics. To get a non-trivial theory one adds a scalar field, the dilaton ϕ_d , which multiplies the curvature. The Jackiw–Teitelboim action is:

$$I = - (1/2) \int d^2x \sqrt{g} \phi_d (R + 2) - \int \partial \sqrt{h} \phi_b (K - 1)$$

The dilaton appears linearly, so integrating it out imposes a constraint rather than producing dynamics:

$$R + 2 = 0 \implies \text{the bulk is rigidly } AdS_2$$

There is no bulk graviton, no local degrees of freedom, and no choice of geometry. Every solution has the same bulk. All of the physics is therefore in the boundary term, which is the Gibbons–Hawking–York term evaluated on the wiggling boundary curve.

The boundary particle and the emergence of the Schwarzian

The only remaining freedom is the shape of the boundary curve in rigid AdS_2 — where the cut-off surface sits, parametrized by proper boundary time. Evaluating the extrinsic curvature K for a curve at large but finite radial cutoff and expanding gives, after the divergent piece is cancelled by the counterterm:

$$I_{bdy} = -C \int du \text{Sch}(t(u), u)$$

The same Schwarzian action, obtained by a completely different route. On the SYK side it came from breaking an emergent symmetry of a disordered many-body system; on the JT side it came from the shape of a boundary curve in a rigid geometry. That these coincide is the SYK/JT correspondence.

The dictionary — IMPORTED, not measured

The map relating the SYK parameters (J , N , α_S , β) to the JT quantities (C , the horizon location, the geometric barcode scales) is the large- N Maldacena–Stanford dictionary. In this project it is used as an imported calibration taken from

the literature. We did not derive it and we did not measure it. This tag is carried through every downstream claim in Part IX. Treating an imported relation as a measured result is precisely the error that produced the Rung 1–2 tautology.

3.6 The thermofield double state

The object at the heart of ER=EPR. Take two identical copies of a quantum system, label them L and R, and form:

$$|TFD(\beta)\rangle = Z(\beta)^{-1/2} \sum_n e^{-\beta E_n/2} |n\rangle_L \otimes |n\rangle_R$$

Three facts define its role:

- 4. Tracing out either side gives exactly the thermal density matrix of the other: $\text{Tr}_L |TFD\rangle\langle TFD| = e^{-\beta H_R} / Z$. A single-sided observer sees a thermal state at temperature $1/\beta$ and cannot tell it is part of a pure state.
- 5. The entanglement entropy between L and R is the thermodynamic entropy $S(\beta)$ of one copy. Entanglement and thermodynamics are the same number here.
- 6. Correlators of operators inserted on the two sides are analytic continuations of Euclidean correlators around the thermal circle. The thermal circle of circumference β is the geometric arena, and this is why our filtration is built on it.

Why this state is the ER=EPR test case

Maldacena's eternal black hole in AdS has two asymptotic boundaries connected through a non-traversable Einstein–Rosen bridge. Its dual is exactly $|TFD(\beta)\rangle$ on two boundary copies. The entanglement between L and R is the wormhole. Temperature controls both: raising the temperature raises the entanglement and, on the gravity side, changes the horizon and the geometry of the bridge. If barcodes of entanglement are barcodes of geometry anywhere, this is where.

3.7 The full physical pipeline

Stage	Boundary (SYK) side	Bulk (JT) side
Input	N, J, β ; disorder realization of J_{ijkl}	C (Schwarzian coupling), β
Core computation	Exact diagonalization ($N \leq 18$) or Schwinger–Dyson iteration at large N	Schwarzian correlator, MTV 1705.08408
Observable	$G(\tau)$ on discretized thermal circle, $N_\tau = 16$	Geodesic / correlator data on the same τ grid
Metric	$d(\tau_i, \tau_j) = \text{monotone decreasing function of } G $	Same functional form, bulk-derived G
Topology	VR filtration $\rightarrow$ persistence module F_{bdy}	VR filtration $\rightarrow$ persistence module F_{bulk}
Comparison	$\phi : F_{\text{bdy}} \Rightarrow F_{\text{bulk}}$; Wasserstein / bottleneck distance; naturality check	

The critical design decision is the one in the 'Observable' row: both sides are indexed by Euclidean time τ on the same grid. Part IV explains why anything else is incoherent.

Part IV — The Bridge: Constructing ϕ

Two well-understood theories and a well-understood invariant do not automatically make a comparison. The construction that joins them is where every subtle error in this project lived, and where the real design work happened.

4.1 The index mismatch, and why it is fatal if unnoticed

The SYK model has a natural index set: the fermion flavor label $i \in \{1, \dots, N\}$. It is tempting to build the point cloud out of flavors — treat each Majorana as a point, use the correlation matrix $\langle \chi_i \chi_j \rangle$ as a distance, and compute persistence. Several early sketches of this project did exactly that.

JT gravity has a natural index set too: radial position in the bulk, the holographic coordinate. It is equally tempting to build the bulk point cloud out of radial slices.

The mismatch — a structural finding

Fermion flavor index and bulk radial position are NOT related to each other by the holographic dictionary. There is no entry in the dictionary that maps flavor i to radius z . A comparison indexed one way on the boundary and the other way in the bulk is comparing objects that are not images of each other under any known map, and the resulting agreement or disagreement means nothing.

Worse, the flavor construction has a second defect. In the disorder-averaged large- N theory the two-point function is flavor-diagonal, $\langle \chi_i \chi_j \rangle \propto \delta_{ij} G(\tau)$. The flavor-indexed distance matrix is therefore structureless — every off-diagonal entry is the same — and the resulting point cloud is a regular simplex, whose Vietoris–Rips filtration has a completely trivial and physics-independent barcode. It would have produced clean, stable, meaningless output.

The resolution: τ -indexing on both sides

The only index shared by both descriptions is Euclidean time on the thermal circle. On the boundary side, τ parametrizes the thermal circle of circumference β . On the bulk side, τ parametrizes the boundary curve of the JT geometry — the same circle, since the Euclidean black hole is a disk whose boundary is that circle. This is a genuine identification supplied by the correspondence, not an assumed one.

So both point clouds are built on the same object: $N_\tau = 16$ points equally spaced on the thermal circle. This is the construction that makes the comparison well posed. It is also, as Part VI shows, the construction that makes the comparison structurally impoverished — the same decision that makes it coherent is the one that caps its information content.

4.2 From correlator to metric

A persistence computation needs a distance. The correlator $G(\tau_i, \tau_j)$ is a similarity: large when points are strongly correlated, small when they are not. A metric must be the reverse — small for similar, large for dissimilar — and must be monotone so that the filtration nests correctly.

We therefore take:

$$d(\tau_i, \tau_j) = \Phi(| G(\tau_i, \tau_j) |)$$

where Φ is a fixed, strictly decreasing function. The requirements on Φ are:

- Strictly decreasing, so that stronger correlation means smaller distance.
- Applied identically on both sides, so that any difference in the resulting barcodes is a difference in the physics and not a difference in convention.
- Monotone in the filtration parameter, so that $VR(X, r) \subseteq VR(X, r')$ for $r \leq r'$ — the property Part II identified as non-negotiable.

Note what is deliberately not required: that Φ satisfy the triangle inequality. Vietoris–Rips is defined for arbitrary symmetric dissimilarity matrices, and the persistence module and its barcode exist regardless. Insisting on a true metric would have imposed a constraint the mathematics does not need, and would have been an invented condition of exactly the kind Part VIII catalogues.

The convention-robustness requirement

A conclusion that depends on the choice of Φ is a conclusion about Φ , not about physics. Every claim in Rungs 3 and 4 was therefore re-run under alternative choices of Φ and checked for stability. This is what exposed the '2/16 favored' artifact described in Part VIII: under one convention the result looked like a weak positive, and under the convention sweep it dissolved into UV-crossing noise.

4.3 Definition of ϕ

With both sides producing persistence modules indexed by the same scale parameter, ϕ is defined component-wise. For each scale r , ϕ_r is the linear map

$$\phi_r : F_{\text{bdy}}(r) \rightarrow F_{\text{bulk}}(\lambda(r))$$

where λ is the scale-matching reparametrization supplied by the holographic dictionary — the calibration that says which bulk scale corresponds to which boundary scale. The map ϕ_r sends each persistent homology class of the boundary filtration to the class of the corresponding cycle in the bulk filtration, under the identification of the two τ -grids.

Two things must then be proven, and they are the content of Rung 5:

7. Functoriality / naturality: for every $r \leq r'$, the square $F_{\text{bulk}}(\lambda(r) \leq \lambda(r')) \circ \phi_r = \phi_{r'} \circ F_{\text{bdy}}(r \leq r')$ commutes. This is what makes ϕ a natural transformation rather than an unrelated collection of maps.
8. Invertibility: each ϕ_r is an isomorphism. By the structure theorem this reduces to a statement about barcodes — the interval decompositions must match — which is why the structure theorem is the tool for the invertibility half.

4.4 The tautology trap, stated in general form

This is the most important subsection in Part IV, because it is the error the project came closest to publishing.

Suppose the calibration λ is fixed by demanding that some boundary quantity equal some bulk quantity. Suppose ϕ is then defined using λ . Then the agreement of those quantities is not a test of anything — it is a restatement of how λ was chosen. If λ is fixed by matching entropies, then entropies will match. If λ is fixed by matching a barcode feature, that feature will match. The computation will run, produce a number, and the number will be zero, and the zero will mean nothing.

The general principle — the central methodological finding

Agreement is evidence only when the agreeing things are independent. Before any matching result is interpreted, the construction must be audited for the question: could this agreement have been produced without any physical input? If yes, the agreement is construction-guaranteed and must be quarantined, not reported.

Rungs 1 and 2 failed this audit. ϕ had been defined as the ratio of two quantities that were defined to be equal, so $W = 0$ was algebraically forced. The correct response was not to discard the rungs — they still validate the pipeline — but to relabel them: pipeline validation, not physics result. That relabeling is what made Rung 3 an honest experiment.

How Rung 3 onward avoids the trap

From Rung 3 the calibration is fixed once, in one channel, using the imported large-N dictionary, and is then held fixed while a different, independent quantity is measured. The barcode shape is not used to set the calibration and is therefore free to disagree — and in Rung 4 it does. Specifically, the shape failure of the Schwarzian profile is established independent of calibration: no choice of the constant can repair it. That independence is what makes the Rung 4 negative a real measurement.

4.5 The Jacobian question — closed

Converting between the boundary parametrization and the variables in which the Schwarzian correlator is evaluated introduces a Jacobian factor. In the implementation this appears as a single additive constant in log space:

```
LOG_JACOBIAN = -log(2) # schwarzian.py, line 68
```

This constant was the subject of an extended and initially unsatisfying exchange, recorded in Part VIII. It was ultimately settled empirically by a direct A/B test: the pipeline was run with and without the factor and the outputs compared against the independently-known limiting behaviour. The result was unambiguous.

Status: CLOSED — resolved by A/B test

The Jacobian is correct as implemented. This question is settled and is not to be re-raised. It is recorded here so that a future reader — or a future collaborator inheriting the code — does not spend another cycle on it.

Part V — The Experimental Record, Rung by Rung

Every experiment run in this project, with its design, its output, and the interpretation that survived audit. Where an initial interpretation did not survive, both the initial and the corrected reading are given.

5.1 Rung 1 — Two entangled qubits

Design

The smallest system in which entanglement exists. Take the Bell state:

$$|\Phi^+\rangle = (|00\rangle + |11\rangle) / \sqrt{2}$$

Build the correlation matrix, convert it to a distance matrix by the Φ prescription, compute the Vietoris–Rips persistence, and compare to the barcode of the corresponding minimal geometry. Compute the Wasserstein distance between the two diagrams.

Result

```
W(F_bdy, F_bulk) = 0.000 (exactly)
```

Initial reading

This looked like an immediate and total confirmation. Two independent computations — one quantum-mechanical, one geometric — landing on exactly the same barcode, at the very first attempt.

Audited reading — the reading that stands

Rung 1 is pipeline validation, NOT a physics result

The vanishing Wasserstein distance was algebraically guaranteed. ϕ was constructed as the ratio of two quantities that had themselves been defined to be equal, so $W = 0$ follows from the definitions before any state is specified. Changing the physics would not have changed the answer. The Ryu–Takayanagi relation we believed we were confirming was an input to the construction, not an output of it.

What Rung 1 does establish, legitimately: the code runs end to end, the distance conversion is monotone, the persistence computation returns well-formed barcodes, and the Wasserstein comparison is correctly implemented. That is genuine and necessary. It is simply not a statement about ER=EPR.

5.2 Rung 2 — Thermofield double at varying temperature

Design

Extend to $|\text{TFD}(\beta)\rangle$ and sweep β . The hypothesis is that as temperature varies, the entanglement structure changes, the dual geometry changes, and the two barcodes should track each other across the sweep. Temperature dependence is a much richer test than a single state: a construction can accidentally match at one point but is unlikely to accidentally match along a curve.

Result

Barcodes tracked across the full β sweep. Wasserstein distance remained at or near zero throughout.

Audited reading

Rung 2 inherits the Rung 1 tautology in full. The same ϕ , defined by the same ratio, was used. The tracking across β is impressive-looking and completely uninformative: because the calibration was itself a function of β fixed by the matching condition, the match was maintained by construction at every temperature. A curve of guaranteed agreements is not more evidence than a single guaranteed agreement.

Rung 2's genuine contribution is that it introduced the thermal circle as the geometric arena and forced the discretization decision — N_τ points on a circle of circumference β — that everything above depends on.

Why Rungs 1–2 were kept in the record

There was an obvious temptation to quietly drop two rungs that turned out to be tautological. They are retained, clearly labelled, for two reasons. First, they are honest pipeline validation and the pipeline needed validating. Second, the diagnosis of why they were tautological is itself the project's central methodological result, and it cannot be stated without the example.

5.3 Rung 3 — Full SYK versus JT gravity

Design

The first rung where the answer is not fixed by construction. On the boundary side: exact diagonalization of the SYK Hamiltonian, computation of $G(\tau)$ on the discretized thermal circle, with disorder averaging over realizations of J_{ijkl} . On the bulk side: the JT gravity geometry at the corresponding temperature, with the Schwarzian correlator taken from MTV 1705.08408, evaluated on the same τ grid. Both sides then pushed through the identical persistence pipeline.

The structural finding

Rung 3's most important output was not a number. It was the discovery of the index mismatch described in Section 4.1: that the flavor-indexed construction being used was comparing incommensurable objects, and that both sides had to be re-indexed by τ . This was a redesign, not a bug fix, and it is the reason Rung 3 took as long as it did.

Result after re-indexing

With τ -indexing on both sides, genuine non-trivial topological features appear for the first time — H_1 bars with birth and death scales that depend on the physics rather than on the construction. The comparison becomes meaningful because the barcodes are now free to disagree.

What Rung 3 did not settle

Rung 3 established that a well-posed comparison exists and that it produces non-trivial output. It did not establish that the comparison succeeds. Any agreement at Rung 3 is at the level of a single calibrated channel, and the whole lesson of Rungs 1–2 is that a single calibrated channel proves nothing. Rung 4 exists to test an uncalibrated channel.

5.4 Rung 4 — Finite-N corrections: the honest negative

Committed as an honest negative

Commits 3ee3035 and a2204e5, with closure text, verbatim rescore output, and a proven-versus-conjectured ledger in rung4_results.txt. Nothing in Rung 4 was reported that was not computed, and the verbatim output is committed alongside the interpretation so that a reader can check the interpretation against the numbers.

Design

If the SYK/JT correspondence holds and ϕ is approaching a natural isomorphism, then at finite N the barcodes should differ by an amount that behaves like a quantum correction: suppressed by a definite power of $1/N$, with a calibration constant that is genuinely constant — independent of temperature and of which barcode channel is used to extract it. This is a sharp, falsifiable prediction, and it is not the channel used to fix the calibration.

The test therefore fits a calibration constant C_{fit} from the barcode data at each N and each β , and asks three questions: is it independent of β , does it scale with N as a power law, and is it the same constant when extracted from different channels?

Result 1 — C_{fit} is β -dependent

Spread of C_{fit} across beta, at fixed N : 29% to 158%

A calibration constant that varies by up to 158 % across temperature at a single system size is not a constant. If the correspondence held with a well-defined calibration, this spread should shrink toward zero. It does not.

Result 2 — the power-law fit is uninformative

Fit form: $C_{\text{fit}} = a * N^b$ Result: b carries large uncertainty

The $1/N$ -suppression prediction requires a definite exponent. The fitted exponent b has an uncertainty large enough that it does not distinguish the predicted scaling from alternatives. We report this as an uninformative fit rather than as a measured exponent. Quoting a central value for b without its uncertainty would have been exactly the kind of numeric assertion this project refuses to make.

Result 3 — the two channels disagree

Ratio $C_{\text{fit}}(\hat{G}(1)) / C_{\text{fit}}(\hat{G}(8))$ ranges from 1.47 to 7.81

This is the sharpest single number in the project. The structure theorem of Part VI shows that the entire barcode reduces to two scalar functionals of the correlator, $\hat{G}(1)$ and $\hat{G}(8)$. Each channel independently determines a calibration constant. If a single natural isomorphism existed, the two constants would be the same constant, and the ratio would be 1. It ranges up to 7.81. No single calibration can satisfy both channels simultaneously.

Result 4 — shape failure, independent of calibration

The strongest result, because it cannot be argued away by adjusting a constant. The Schwarzian profile predicted on the bulk side and the profile measured on the boundary side differ in shape, not merely in normalization. A calibration constant is a multiplicative rescaling; a rescaling cannot change a shape. Therefore:

The shape failure is established independent of calibration — MEASURED

At every system size accessible to exact diagonalization ($N \leq 18$, Hilbert space dimension $2^9 = 512$), the barcode profile does not match the Schwarzian prediction in shape. No choice of C , and no choice of the distance convention Φ , repairs this. This is the load-bearing negative result of the project.

What Rung 4 does NOT claim

Stating the boundaries of the result precisely, because the boundaries are as important as the result:

- It does not claim ϕ fails in the large- N limit. Every measurement here is at $N \leq 18$. The large- N behaviour is not accessible to exact diagonalization and we did not measure it.
- It does not claim $ER=EPR$ is false. It claims that this particular construction of ϕ , on this particular filtration, is not a natural isomorphism in the regime measured.
- It does not claim the deviations are not quantum corrections. It claims that the specific quantitative signature of quantum corrections — a β -independent constant with a definite N -scaling — is not observed.

The reachability correction

An earlier and stronger claim was drafted here: that no value of C could reach the SYK data at all. That claim was wrong, and its refutation is recorded in Part VIII. It was extracted from a sweep at a single β ; the $\beta = 40$ row exhibits an interior minimum, which directly contradicts it. The corrected statement — that no single C works across all β — is weaker, and it is the one that is true.

5.5 Rung 5 — status

Rung 5 is Naman's, and it is in progress. The structure theorem of Part VI and the corollary of Part VII are drafted. The outstanding item is the functoriality proof: showing that every naturality square commutes for all morphisms in the filtration poset, rather than checking components pointwise. Pointwise invertibility without naturality would not be a natural isomorphism, and the distinction is the entire reason the project is stated categorically.

Part VI — The Structure Theorem

The negative result of Rung 4 is not an accident of resolution or of system size. It has a proof. This part gives it, and the proof is the most valuable object the project produced.

6.1 The setup

Both persistence modules are built on the same object: $N_\tau = 16$ points equally spaced on the Euclidean thermal circle of circumference β . Label them $\tau_0, \dots, \tau_{15}$ with $\tau_i = i\beta / 16$.

The correlator on the thermal circle depends on the two arguments only through their separation, and is symmetric under reflection:

$$G(\tau_i, \tau_j) = G(|i - j| \bmod 16) \quad \text{and} \quad G(s) = G(16 - s)$$

This is not an approximation. It follows directly from time-translation invariance on the thermal circle and from the $\tau \rightarrow \beta - \tau$ symmetry of the finite-temperature correlator derived in Section 3.3. Both properties hold exactly for the conformal solution and exactly for the exact-diagonalization data.

Since the distance is $d = \Phi(|G|)$ with Φ a fixed function, the distance matrix inherits both properties:

$$d(\tau_i, \tau_j) = d(|i - j| \bmod 16) =: d(s), \quad d(s) = d(16 - s)$$

The distance matrix is circulant and symmetric

A 16×16 circulant symmetric matrix is determined entirely by the 8 numbers $d(1), d(2), \dots, d(8)$. Separations 9 through 15 are mirror images of 7 through 1, and separation 0 is the diagonal. The entire boundary dataset — regardless of β , regardless of N , regardless of the disorder realization — is 8 real numbers.

6.2 The reduction

Step 1 — the filtration is a filtration of circulant graphs

At scale r , the graph G_r has an edge between τ_i and τ_j exactly when $d(|i - j|) \leq r$. Because d depends only on separation, the set of separations admitted at scale r is determined by which of the eight values $d(1), \dots, d(8)$ fall below r . Hence G_r is a circulant graph $C_{16}(S_r)$ where $S_r \subseteq \{1, \dots, 8\}$ is the admitted separation set.

Because the filtration is monotone in r , S_r grows monotonically as r increases. And because the eight values $d(s)$ are simply real numbers, the entire filtration is determined by their ordering and their values — nothing else about the correlator enters.

Step 2 — Adamaszek–Adams applies

The Vietoris–Rips complex is the clique complex of G_r . For circulant graphs on a cycle, the Adamaszek–Adams classification of Section 2.7 applies: the homotopy type of the clique complex of $C_{16}(S)$ is determined by S , and passes through the rigid sequence $S^1, S^3, S^5, \dots$ as the admitted separation set grows. It is never anything else.

Therefore the homotopy type at every scale is a function of S_r alone, and hence a function of the 8-vector ($d(1)$, ..., $d(8)$) alone.

Step 3 — the barcode is a function of the 8-vector

Structure theorem — PROVEN

Let the filtration be induced by a monotone decreasing function of a correlator on a discretized thermal circle with $N_\tau = 16$ points. Then the persistence module in every homological degree is completely determined by the ordered 8-vector ($d(1)$, ..., $d(8)$). No other feature of the correlator — no other moment, no phase information, no higher-point structure — enters the barcode at any scale.

Step 4 — the collapse to two scalars

The structure theorem says the barcode depends on at most 8 numbers. Empirically, and in accordance with the Adamaszek–Adams threshold structure, it depends on far fewer. The persistent H_1 bar — the only bar carrying physical content — has:

- A birth determined by when the cycle first closes, which is controlled by the nearest-neighbour separation, $\hat{G}(1)$.
- A death determined by when the complex jumps from S^1 to S^3 , which is controlled by the long-separation entry, $\hat{G}(8)$ — the antipodal correlator on the thermal circle.

Sensitivity analysis on the computed barcodes gives the weighting of these two channels as approximately 23 : 1. The barcode is overwhelmingly a function of the nearest-neighbour correlator, with the antipodal correlator supplying a subleading but non-negligible correction, and the remaining six entries contributing essentially nothing.

Epistemic status of the collapse

The reduction to the 8-vector is PROVEN. The further collapse onto $\hat{G}(1)$ and $\hat{G}(8)$ with a ~23:1 weighting is MEASURED, and its exact form follows from pinning the death-controlling separation index to the corresponding Adamaszek–Adams threshold — an identification that is drafted but not yet written as a proof. This is listed as an open item in Part IX rather than asserted.

6.3 The UV-localization mechanism

The 23 : 1 weighting has a physical reading, and it explains why the construction is structurally biased against detecting the physics it was built to detect.

The nearest-neighbour separation on the thermal circle is the shortest Euclidean time separation on the grid — it probes the ultraviolet. The conformal correlator diverges as $\tau \rightarrow 0$, so $|G(1)|$ is the largest entry in the correlator by a wide margin, which is exactly why it dominates the distance matrix and therefore the filtration.

But the physics the project is trying to detect — the Schwarzian mode, the reparametrization dynamics, the geometry of the wormhole — is infrared physics. It lives at separations of order β , not at the grid spacing. The barcode is dominated by the one channel where the interesting physics is weakest.

UV-localization, stated as a design conclusion

The single-boundary circulant construction localizes the entire barcode onto the UV end of the correlator, where the Schwarzian contribution is subleading. The construction is not merely insensitive to the physics of interest — it is structurally biased toward the regime where that physics is smallest. This is why the finite- N deviations do not behave like clean quantum corrections, and it is a property of the construction, not of the physics.

6.4 Why this is a better result than the isomorphism would have been

Suppose ϕ had turned out to be a natural isomorphism. Given the structure theorem, what would that have meant? It would have meant that two 8-vectors, each further collapsing to two numbers, happened to agree — after a calibration constant had been fitted. That is a far weaker statement than it sounds, and given the tautology diagnosis of Part IV, it would have required extraordinary care to establish that it was not construction-guaranteed.

What we have instead is a theorem that holds for every construction of this type, not just ours: any single-boundary circulant thermal-circle filtration carries at most 8 real numbers of topological information, effectively 2. That statement constrains all future attempts. It converts 'our construction did not work' into 'no construction of this class can work, and here is precisely what a working construction must do differently.'

Part VII — Rung 5: The Categorical Statement

The target theorem, its corollary, and the falsifiable criterion. This is Naman's section and the mathematical payload of the paper.

7.1 The main theorem

Theorem (structure of circulant thermal-circle persistence)

Let $\beta > 0$ and let G be a correlator on the discretized thermal circle with N_τ points satisfying time-translation invariance and reflection symmetry. Let Φ be a strictly decreasing function and $d = \Phi(|G|)$. Then the persistence module $F = H_\bullet \circ VR \circ d$, as a functor $(\mathbb{R}, \leq) \rightarrow \text{Vect}_k$, factors through the finite-dimensional space of ordered separation vectors $(d(1), \dots, d(\lfloor N_\tau/2 \rfloor))$. In particular, for $N_\tau = 16$ the functor is determined by 8 real parameters.

7.2 The corollary — the falsifiable criterion

This is the operational content and the reason the negative result is publishable.

Corollary (criterion for natural isomorphism)

With F_{bdy} and F_{bulk} both of the above form, a natural transformation $\phi : F_{\text{bdy}} \Rightarrow F_{\text{bulk}}$ is a natural isomorphism if and only if the boundary and bulk separation vectors agree up to a single common reparametrization λ of the scale axis. Equivalently: the calibration constant extracted from each independent barcode channel must be the same constant, and the profile shapes must agree, not merely the normalizations.

Application of the criterion to the measured data

The criterion is checkable, and Rung 4 checked it. It fails on both clauses:

Clause of the criterion	What it requires	What is measured	Verdict
Single common calibration	C_{fit} identical across channels; ratio = 1	$C_{\text{fit}}(\hat{G}1)/C_{\text{fit}}(\hat{G}8)$ ranges 1.47 – 7.81	FAILS
Calibration independent of β	C_{fit} constant across temperature	Spread of 29 % – 158 % at fixed N	FAILS
Profile shapes agree	Schwarzian shape matched, calibration-independent	Shape mismatch at all $N \leq 18$	FAILS

Three independent clauses, three independent failures, at every accessible system size. The criterion is not marginally violated.

7.3 What remains to be proven

- Functoriality. Every naturality square must be shown to commute for all morphisms $r \leq r'$ in the filtration poset. This is the outstanding proof obligation and the current active work.

- Invertibility via the structure theorem. Given functoriality, invertibility of each component reduces by the interval decomposition to agreement of barcodes, which the corollary then characterizes exactly.
- The Adamaszek–Adams threshold identification. Pinning the death-controlling separation index rigorously, which upgrades the two-scalar collapse from MEASURED to PROVEN.

7.4 The reframed paper structure

The negative result forced a restructuring of the paper. The current framing, which is the correct one, presents the work not as a failed isomorphism but as a structure theorem with an application:

9. A structure theorem on circulant thermal-circle filtrations — the general mathematical result of Part VI, which stands independent of any physics.
10. A corollary characterizing exactly when ϕ is a natural isomorphism — the falsifiable criterion of Section 7.2.
11. The SYK/JT comparison as an application, showing the criterion fails at $N \leq 18$, with the full Rung 4 measurement record.
12. A methodological conclusion pointing toward two-boundary constructions as the correct next step.

Under this framing the negative result is not a disappointment reported at the end; it is the worked example that demonstrates the criterion has teeth. A criterion that nothing ever fails is not a criterion.

Part VIII — Errors Caught: The Epistemic Record

Every substantive error made in this project, why it was wrong, and how it was found. This section exists because a research record that reports only successes is not a research record. It is also, candidly, the section that best demonstrates what was actually learned.

8.1 The working method: wall detection

The implementation work runs through Claude Code as an implementation agent. The discipline that governs it is: every plan and every output is pasted into a separate reasoning context and sanity-checked for physics and mathematics before the step is approved. The operative test is a single question:

The test

Can I derive this cold, or did I pattern-match? If the derivation cannot be reproduced from scratch without looking at the output, the step is not approved regardless of how reasonable it looks.

Approval prompts — bash commands, file writes — are reviewed with explicit reasoning about whether to grant one-off or standing approval. They are not rubber-stamped. Standing approval is granted only for operations whose failure mode is recoverable and visible.

The five failure modes watched for on every review

13. Invented or unjustified parameters and conventions presented as physics — regulator choices, prefactor calibrations, normalization conventions that appear without derivation.
14. Scope broadening done quietly. A step that was approved to do X quietly also doing Y.
15. Numeric assertions substituted for numeric verification. Numbers must be shown, not asserted. 'The fit gives approximately the expected value' is not a number.
16. Explanations that answer an adjacent question rather than the one asked. This is the subtlest and hardest to catch.
17. Overgeneralizations from limited parameter sweeps presented as global claims.

8.2 Error 1 — the Rungs 1–2 tautology

Diagnosed in Section 4.4 and Section 5.1. ϕ was defined as the ratio of two quantities that were themselves defined to be equal; $W = 0$ was algebraically guaranteed; the Ryu–Takayanagi relation was an input, not a discovery.

How it was caught: by asking of a perfect result the question 'what would have had to be true for this to come out differently?' and finding that the answer was 'nothing.' A result that cannot fail is not a result.

Severity: highest. Had this not been caught, the project would have reported a spectacular confirmation of ER=EPR that was a restatement of its own definitions.

8.3 Error 2 — the shared-Jacobian cross-check

A cross-check was performed between two calculations that both used the same Jacobian convention. It agreed, and the agreement was initially read as validating the convention.

Why this is wrong: two calculations sharing an input cannot validate that input. The cross-check confirmed internal consistency, which is a real but much weaker property than correctness. Consistency tests and correctness tests are different tests and the difference must be stated when the result is reported.

Resolution: the Jacobian was subsequently settled by an actual A/B test against independently known limiting behaviour, as recorded in Section 4.5.

8.4 Error 3 — the unanswered Jacobian question

The Jacobian question was posed and went unanswered across multiple turns, during which a domain-geometry explanation was supplied instead. The domain-geometry material was correct in itself. It simply was not the answer to the question asked.

The failure mode this exemplifies

Substituting an answer to an adjacent question for an answer to the question asked, without flagging the substitution. The substitution is what makes it an error — a correct answer to a different question, clearly labelled as such, is fine. Unlabelled, it consumes the turn and leaves the original question silently open. This is the hardest of the five failure modes to detect, because the material supplied is genuinely correct.

Detection required tracking the original question across turns rather than evaluating each response on its own merits. This is now an explicit part of the review discipline.

8.5 Error 4 — the '2/16 favored' misreading

A result in which 2 of 16 configurations favored the hypothesis was initially read as a weak positive signal.

It was not. The split table, broken out by configuration, together with the convention-robustness analysis, showed that the 2 favorable cases were UV-crossing artifacts — cases where the ultraviolet end of the two correlators crossed, producing an apparent preference with no physical content. Under alternative choices of the distance convention Φ , the identity of the favorable cases changed. A signal whose membership depends on convention is noise.

Lesson recorded: a weak positive should be broken out and convention-swept before it is called a positive at all. Aggregate counts hide mechanism.

8.6 Error 5 — the $|d|$ crease/kink mechanism

A mechanism was proposed for the observed (s, d) rotation based on a crease or kink in $|d|$ at $d = 0$.

This is straightforwardly wrong. The relevant function $\Gamma(\Delta + i d)$ is analytic at $d = 0$ — there is no crease, no kink, and no non-smooth behaviour there for a mechanism to exploit. The proposed explanation invoked a feature that does not exist.

The correct explanation is domain geometry: the rotation arises from the shape of the integration domain, not from any non-analyticity of the integrand. Detection came from asking directly whether the claimed non-analyticity was actually present, and checking the analytic structure rather than accepting the mechanism because the conclusion was right.

Note the pattern

The conclusion was correct and the mechanism was wrong. This is more dangerous than a wrong conclusion, because the correct conclusion suppresses scrutiny of the reasoning that produced it. Right answer, wrong reason is a failure state, not a partial success.

8.7 Error 6 — the reachability overgeneralization

The claim 'no C can reach SYK' was made and drafted into the Rung 4 conclusions. It was extracted from a sweep at a single value of β .

It is false. The $\beta = 40$ row exhibits an interior minimum, which directly refutes it: at that temperature there is a value of C that reaches. The corrected and true statement is weaker: no single C works across all β simultaneously.

Rule adopted

Reachability claims require full β coverage. A claim quantified over all values of a parameter cannot be established from a sweep at one value of that parameter, no matter how clean the sweep looks.

8.8 Error 7 — the peak-location estimate

An analytic estimate placed a peak at $k \approx 2500$. Direct computation put it at $k \approx 500\text{--}600$ — the estimate was high by roughly a factor of four.

The lesson is not that the estimate was bad; order-of-magnitude estimates are supposed to be rough. The lesson is that it was carried forward as though it were computed. An estimate and a computation are different objects and must be labelled differently in every downstream use. This is the 'numbers must be shown, not asserted' rule applied to one's own analytics rather than to the implementation agent's output.

8.9 Error 8 — the wrong proofs of the 1D lemma

Two arguments were advanced for the vanishing of H_1 on one-dimensional point clouds: a curvature argument and a triangle-filling argument. Both are wrong, for the reasons given in Section 2.8. The correct proof is via chordality of indifference graphs and collapsibility of chordal clique complexes.

Recorded because the wrong arguments were persuasive. Curvature in particular sounds like exactly the right kind of reason, and is not a simplicial notion at all.

8.10 Error 9 — the index mismatch

Covered in Section 4.1. Fermion flavor indices and bulk radial position indices were being implicitly identified. They are not related by the holographic dictionary. Both sides must be τ -indexed.

This was the most expensive error in engineering terms — it forced a redesign of Rung 3 rather than a patch. It was also the most productive, because the τ -indexed construction is what made the structure theorem visible.

8.11 Summary table

#	Error	Type	How caught	Status
1	Rungs 1–2 tautology	Construction-guaranteed agreement	Asking what could have made the result differ	Corrected; rungs relabelled
2	Shared-Jacobian cross-check	Consistency read as correctness	Noting the shared input	Superseded by A/B test
3	Unanswered Jacobian question	Adjacent-question substitution	Cross-turn question tracking	Closed by A/B test
4	'2/16 favored'	Noise read as weak signal	Split table + convention sweep	Corrected
5	d crease/kink	Wrong mechanism, right conclusion	Checking claimed non-analyticity	Corrected — domain geometry
6	'No C can reach SYK'	Overgeneralization from one β	$\beta = 40$ interior minimum	Corrected — weaker claim
7	$k \approx 2500$ peak estimate	Estimate used as computation	Direct computation: $k \approx 500$ –600	Corrected
8	1D H_1 curvature argument	Invalid proof	Chordality gives the real proof	Corrected
9	Flavor vs radial indexing	Structural incommensurability	Dictionary audit	Corrected — τ -indexing

Part IX — The Proven-vs-Conjectured Ledger

Line-by-line status of every substantive claim. Four tags are used: PROVEN (a mathematical proof exists), MEASURED (computed, with the numbers committed), IMPORTED (taken from the literature and used as calibration), CONJECTURED (believed, not established). Nothing in this project is reported without one of these tags.

9.1 Mathematical claims

Claim	Status	Basis
Persistence modules are functors $(\mathbb{R}, \leq) \rightarrow \text{Vect}_k$	PROVEN	Definition; functoriality of $H_\bullet$ on inclusions
Every pfd persistence module decomposes into intervals	PROVEN	Crawley-Boevey structure theorem
$d_I = d_B$ (isometry theorem)	PROVEN	Standard; licenses reading numerics categorically
Persistent homology is 1-Lipschitz in the filtration	PROVEN	Stability theorem
$VR(S^1; r)$ passes through odd spheres only	PROVEN	Adamaszek–Adams
1D point clouds have $H_1 = 0$ at every scale	PROVEN	Chordality of indifference graphs; collapsibility
Circulant thermal-circle barcode determined by 8-vector	PROVEN	Structure theorem, Part VI
Barcode collapses onto $\hat{G}(1)$, $\hat{G}(8)$ at $\sim 23:1$	MEASURED	Sensitivity analysis; threshold proof outstanding
ϕ is a natural transformation (naturality squares commute)	CONJECTURED	Rung 5 open item — active work
Corollary criterion for natural isomorphism	PROVEN	Given the structure theorem, Part VII

9.2 Physical claims

Claim	Status	Basis
SYK conformal solution $G_\beta(\tau)$, $\Delta = 1/4$	IMPORTED	Standard large-N SYK literature
Schwarzian governs SYK low-energy sector	IMPORTED	Maldacena–Stanford
JT gravity boundary dynamics is Schwarzian	IMPORTED	Standard JT literature
Large-N SYK/JT dictionary (parameter map)	IMPORTED	Maldacena–Stanford — calibration, NOT a measured result
Schwarzian correlator profile used on bulk side	IMPORTED	MTV 1705.08408
TFD is dual to the eternal AdS black hole	IMPORTED	Maldacena; Maldacena–Susskind

Claim	Status	Basis
LOG_JACOBIAN = $-\log(2)$ is correct as implemented	MEASURED	A/B test against known limits — CLOSED

9.3 Measured results (Rung 4)

Result	Status	Value / commit
C_fit is β -dependent at fixed N	MEASURED	Spread 29 % – 158 %; commits 3ee3035, a2204e5
Power-law fit $C_fit = a \cdot N^b$	MEASURED	b carries large uncertainty — uninformative
Two-channel calibration disagreement	MEASURED	$C_fit(\hat{G}1)/C_fit(\hat{G}8) \in [1.47, 7.81]$
Schwarzian shape failure at $N \leq 18$	MEASURED	Independent of calibration and of convention Φ
$\beta = 40$ row has an interior minimum	MEASURED	Refutes the 'no C can reach' claim
Peak location	MEASURED	$k \approx 500\text{--}600$ (analytic estimate of 2500 was $\sim 4\times$ high)
ϕ fails to be a natural isomorphism at $N \leq 18$	MEASURED	Follows from the criterion + the three failures above
ϕ fails in the large-N limit	NOT CLAIMED	Inaccessible to exact diagonalization; not measured
ER=EPR is false	NOT CLAIMED	Out of scope; this tests one construction, not the conjecture

On the last two rows

These are listed deliberately. The most common way a negative result is overclaimed is by quiet expansion of its scope between the results section and the abstract. Writing the non-claims into the ledger makes that expansion visible if it is ever attempted.

Part X — Open Problems and the Path Forward

What the structure theorem tells us to do next. The negative result is directional, not terminal — it names the defect precisely enough to specify the fix.

10.1 The diagnosis, restated as a design requirement

The obstruction is not resolution, not system size, and not numerical precision. It is that a single-boundary circulant thermal-circle filtration carries at most 8 real numbers of topological information, effectively 2, localized at the ultraviolet end where the Schwarzian physics is weakest. Any fix must break the circulant symmetry that causes the collapse.

10.2 Two-boundary constructions — the indicated next step

The thermofield double state has two boundaries, L and R. The construction used throughout Rungs 1–4 builds its filtration on one of them. This is the source of the problem: a single thermal circle is circulant, and circulant is what collapses.

A two-boundary construction builds the point cloud on both copies, with cross-correlators $\langle \chi_L(\tau_i) \chi_R(\tau_j) \rangle$ supplying the off-block distances. Three reasons this is the right move:

18. It breaks the circulant symmetry. The full distance matrix is no longer a function of a single cyclic separation, so the structure theorem of Part VI does not apply and the barcode is not confined to 8 parameters.
19. It probes the wormhole directly. The L–R cross-correlator is the quantity that encodes the Einstein–Rosen bridge. A single-boundary construction never touches it — which, in retrospect, is a striking omission for a project about ER=EPR.
20. It moves weight into the infrared. The L–R correlator at matched times is an infrared quantity, precisely where the Schwarzian lives, correcting the UV-localization identified in Section 6.3.

What must be checked before Rung 6 is trusted

- Monotonicity of the combined distance matrix. Mixing intra-boundary and inter-boundary distances can break the nesting property that the entire persistence apparatus requires. This must be verified, not assumed.
- The tautology audit. The new construction must be checked for construction-guaranteed agreement before any result from it is interpreted. The same discipline that caught Rungs 1–2 applies to Rung 6 with no exceptions.
- Whether the two-boundary distance matrix has its own hidden symmetry. Breaking circulant symmetry is necessary but not sufficient; a different degeneracy could produce a different collapse.

10.3 Other open directions

- Larger N . Exact diagonalization caps out around $N = 18$. Reaching larger N requires either the Schwinger–Dyson route at large N or a sparse/approximate diagonalization scheme. This would test whether the deviations shrink as the correspondence predicts.
- Finer τ discretization. $N_\tau = 16$ gives an 8-vector; $N_\tau = 32$ gives a 16-vector. This does not escape the structure theorem — it only raises the ceiling. Worth quantifying, not worth expecting a qualitative change from.
- Multiparameter persistence. Filtering by scale and temperature simultaneously produces a bipersistence module, which is not classified by a barcode and carries strictly more information. This connects to the secondary project on multiparameter invariants for codimension-two bifurcations and the Bogdanov–Takens normal form, currently tracked but not in active focus.
- Completing the Adamaszek–Adams threshold identification, which upgrades the two-scalar collapse from MEASURED to PROVEN and strengthens Part VI.

10.4 Immediate operational blockers

Highest-priority loose end — unresolved

Push access to github.com/ArjunParadkar/ER-EPR is blocked by a 403. Naman is not listed as a collaborator on the repository. All Rung 4 commits — 3ee3035 and a2204e5, including the closure text, the verbatim rescore output, and `rung4_results.txt` — currently exist only on a local machine with no remote backup. This is a single-point-of-failure risk on the project's most important result and should be resolved before any further work is done.

- Collaborator handoff report for Arjun, covering the Rung 4 closure and the structure-theorem reframe, so that the numerical and geometric pipelines can be aligned to the new framing.
- Rung 5 functoriality proof — the outstanding mathematical obligation.
- Paper integration: Rungs 1–4 are drafted with embedded figures; Part II skeleton is in place; the reframed structure of Section 7.4 needs to be applied throughout.

Part XI — Logistics, Reproducibility, and References

11.1 Division of labor

Naman Guggilam — Research Lead	Arjun Paradkar
Categorical statement of the theorem	SYK numerical implementation
Construction of the natural transformation ϕ	Correlation matrix computation
Proof of functoriality	Persistent homology pipeline
Proof of invertibility via the structure theorem	JT gravity geometry barcodes
All mathematics sections of the paper	Repository maintenance
Structure theorem and falsifiable criterion (Parts VI–VII)	Rescore output and verbatim result logging
Epistemic record and ledger (Parts VIII–IX)	

11.2 Reproducibility

Every numerical claim in this document traces to committed output. The Rung 4 results are at commits 3ee3035 and a2204e5, which include the closure text, the verbatim rescore output, and the proven-versus-conjectured ledger in rung4_results.txt. The Jacobian convention is at schwarzian.py line 68. Interpretations are committed alongside the raw numbers precisely so that a reader can check one against the other.

The standing rule: numbers are shown, not asserted. Any statement of the form 'the fit gives approximately the expected value' is rejected in review and replaced by the value with its uncertainty, or by an explicit statement that the fit is uninformative.

11.3 Timeline

Date	Milestone	Owner
29 July 2026	Pivot: ER=EPR targeted at ISEF as a team entry with Arjun, not STS	Both
Complete	Rungs 1–3, including the τ -indexing redesign	Both
Complete	Rung 4 committed as an honest negative (3ee3035, a2204e5)	Both
In progress	Rung 5 — functoriality proof and categorical write-up	Naman
Immediate	Resolve GitHub 403; deliver handoff report to Arjun	Naman / Arjun
~25 October 2026	arXiv submission target	Both
1 November 2026	MIT / Caltech research portals + Common App, with arXiv link	Naman

ISEF logistics run through the high school club and do not require active attention in the research workflow.

11.4 Key references

- Maldacena & Susskind — Cool horizons for entangled black holes (the ER=EPR conjecture).
- Maldacena — Eternal black holes in anti-de Sitter (the thermofield double / eternal black hole duality).
- Maldacena & Stanford — Remarks on the Sachdev–Ye–Kitaev model (the large-N dictionary, used here as imported calibration).
- Mertens, Turiaci & Verlinde, arXiv:1705.08408 — the Schwarzian correlator used on the bulk side.
- Ryu & Takayanagi — holographic entanglement entropy.
- Adamaszek & Adams — The Vietoris–Rips complexes of a circle (the engine of the structure theorem).
- Crawley-Boevey — Decomposition of pointwise finite-dimensional persistence modules (the structure theorem for barcodes).
- Edelsbrunner & Harer — Computational Topology: An Introduction.
- Oudot — Persistence Theory: From Quiver Representations to Data Analysis.
- Hatcher — Algebraic Topology.
- Riehl — Category Theory in Context.
- Cohen-Steiner, Edelsbrunner & Harer — Stability of persistence diagrams.

11.5 Closing note

The project set out to confirm a natural isomorphism and instead proved a structure theorem that explains why no construction of that class can produce one. Both outcomes were planned for. The reason the second outcome is worth more is that it is falsifiable, checkable, and directional: it specifies exactly what a successful construction must do differently, and Part X names the construction that does it.

The single most transferable thing in this document is not the theorem. It is the audit condition: agreement is evidence only when the agreeing things are independent. Applied early, it cost two rungs. Applied late, it would have cost the paper.